

MAT 3510 — Homework 2

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Due: October 7, 2026

Problem 1 [20 points]

(1) Gaussian elimination via LU decomposition

Task. Write MATLAB code that performs Gaussian elimination with LU decomposition for a linear system $Ax = b$ with $A \in \mathbb{R}^{d \times d}$, $x \in \mathbb{R}^d$, $b \in \mathbb{R}^d$, for general dimension d .

Approach.

Code.

(2) Test example

Task. Test your code on

$$\begin{cases} x_1 - 2x_2 + 3x_3 - 4x_4 = 4 \\ x_2 - x_3 + x_4 = -3 \\ x_1 + 3x_2 + x_4 = 1 \\ -7x_2 + 3x_3 + x_4 = -3 \end{cases} \iff A = \begin{pmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & 1 \\ 0 & -7 & 3 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ -3 \\ 1 \\ -3 \end{pmatrix}.$$

Test script.

Results.